Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087

Experiment No. 04

Aim: To understand Continuous integration, Jenkins installation**.**

LO mapped: LO1, LO3

LO1: Apply Agile methodologies and use Jira for sprint planning, task tracking, and integrating with DevOps workflow

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging, and deployment

Theory: Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

Jenkins is an open-source automation server that facilitates continuous integration and continuous delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supportsa wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

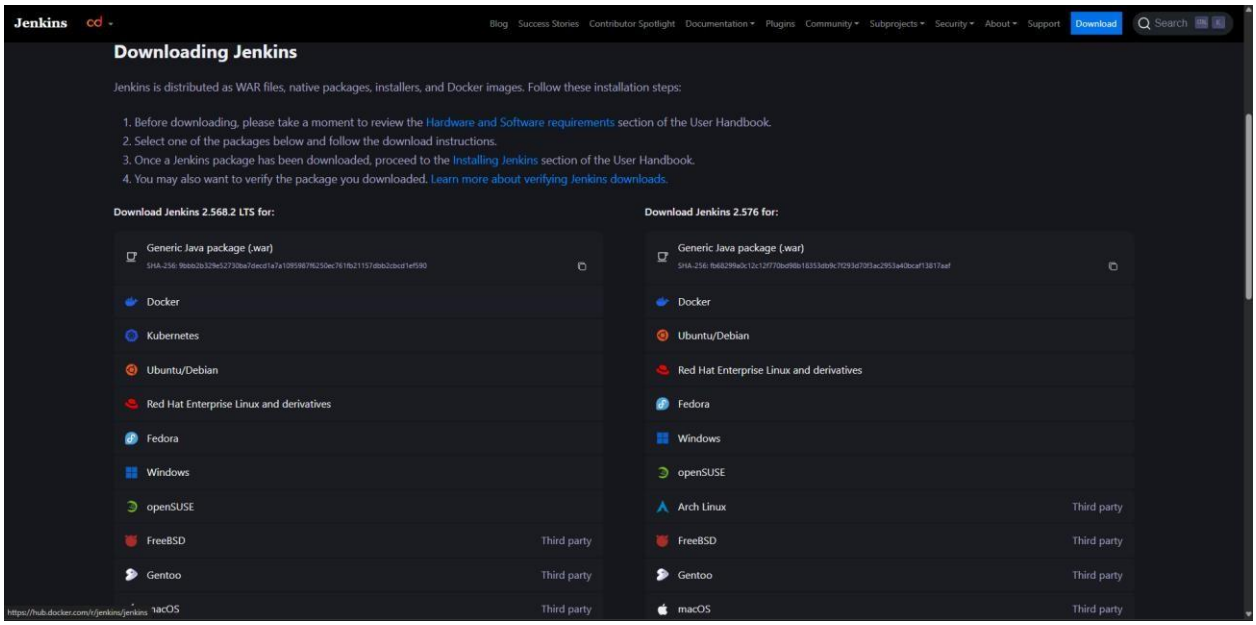
Steps to be followed while installing Jenkins:

1. **Java Installation:** Ensure Java Development Kit (JDK) is installed on your system**.**

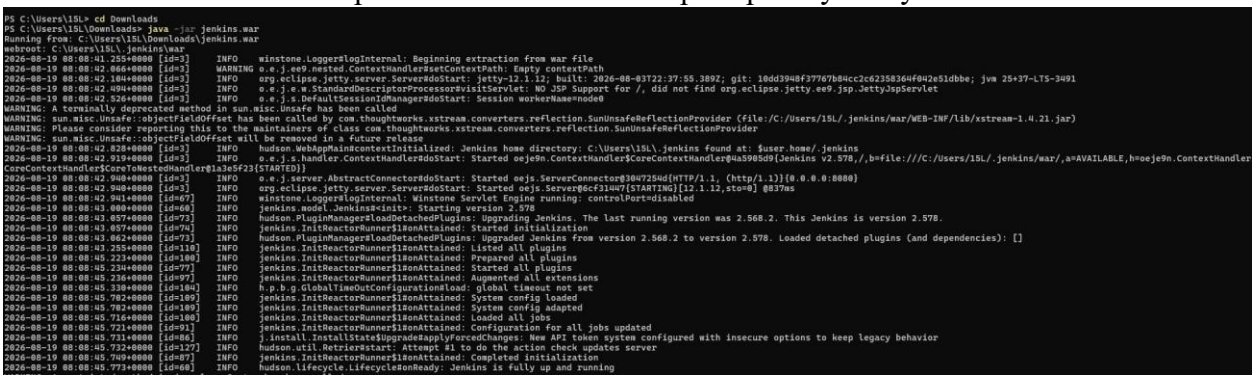
```
C:\Users\15L>java --version
java 25 2025-09-16 LTS
Java(TM) SE Runtime Environment (build 25+37-LTS-3491)
Java HotSpot(TM) 64-Bit Server VM (build 25+37-LTS-3491, mixed mode, sharing)
```

2. **Download Jenkins:** Get the latest Jenkins WAR file from the official website or package manager.

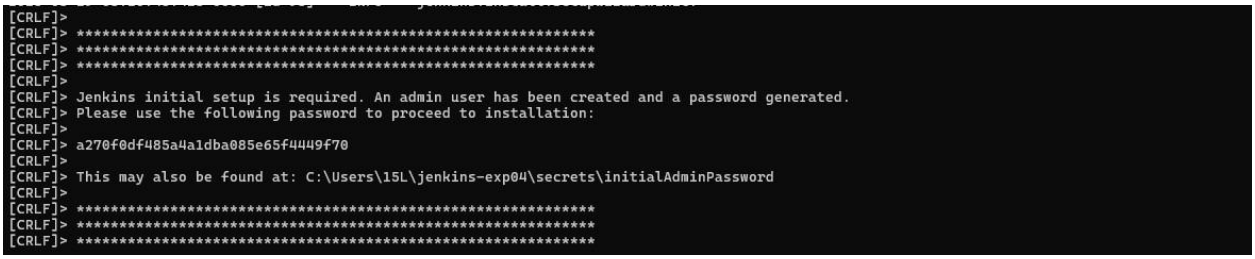
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Batch: T21
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3. **Launch Terminal**: Open a terminal/command prompt on your system.



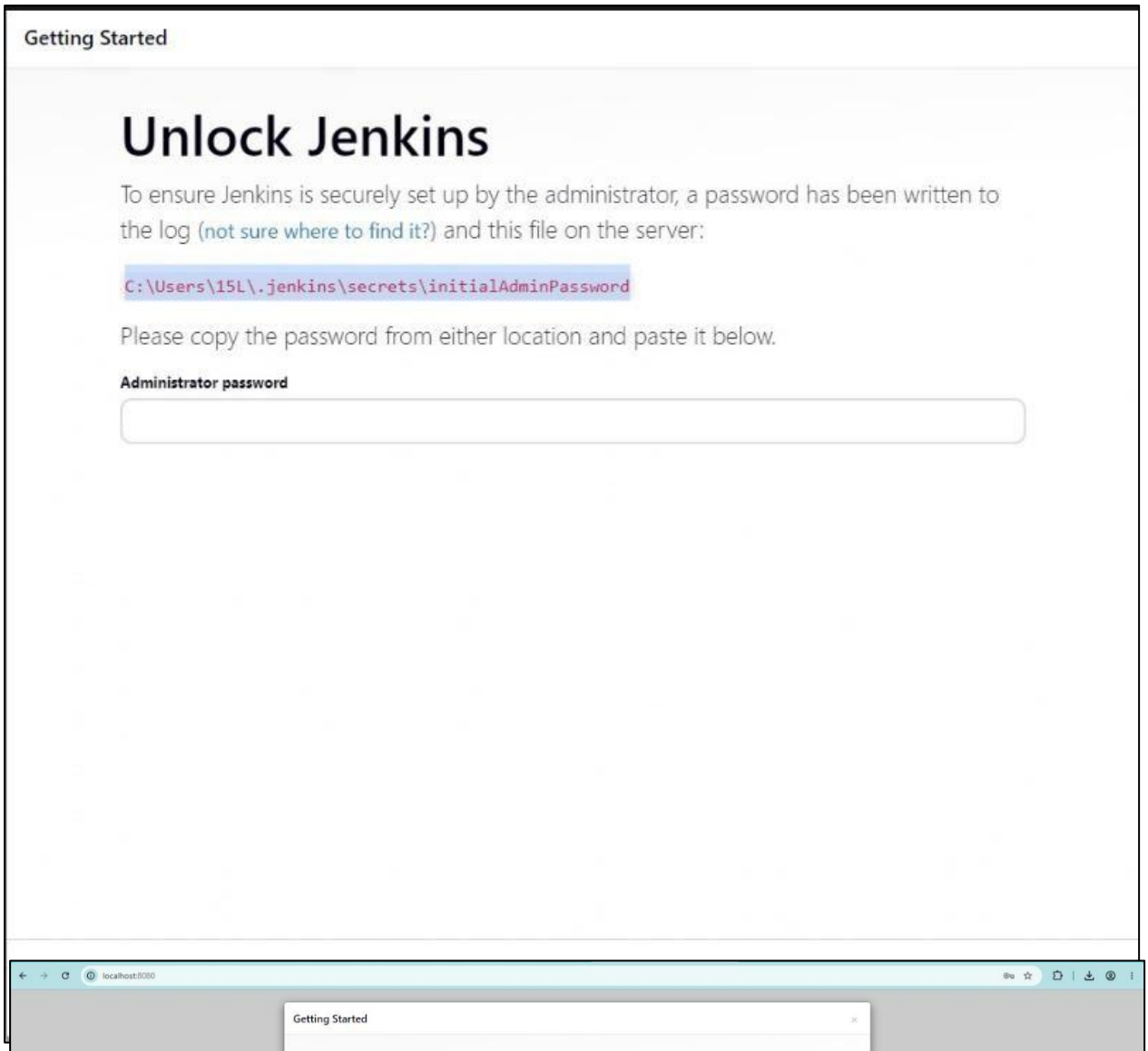
4. **Navigate to Directory**: Use the cd command to navigate to the directory containing the Jenkins WAR file.

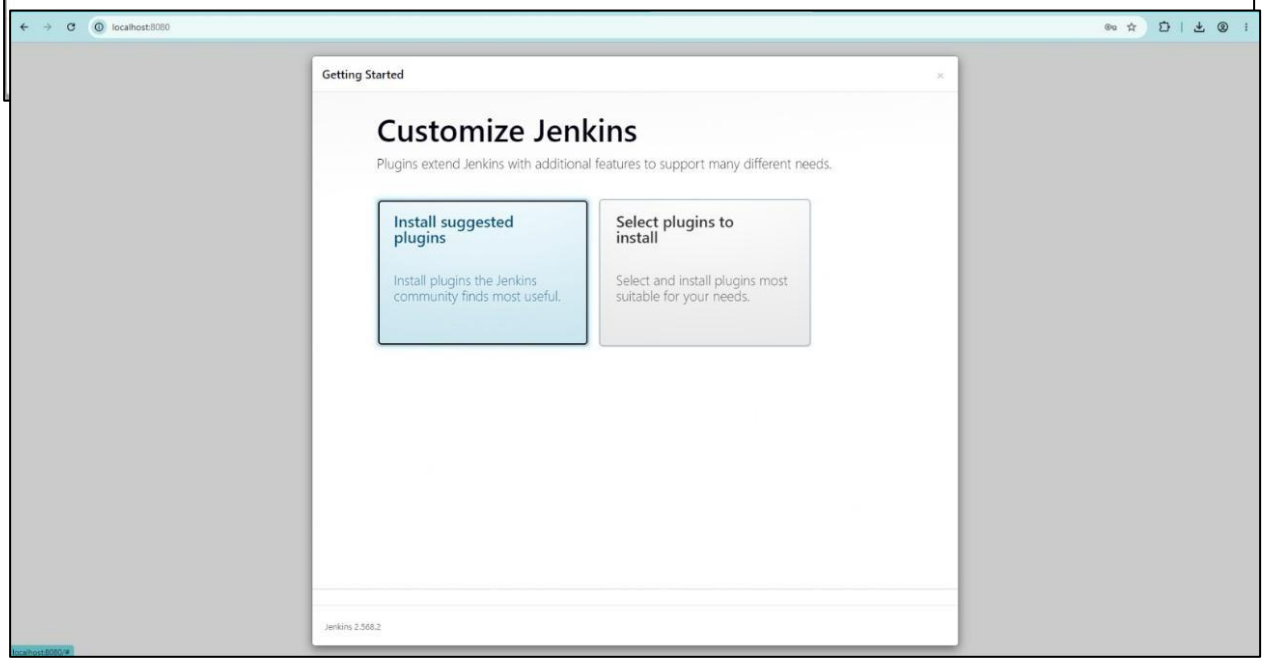


Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087

5. **Start Jenkins**: Run the command java -jar jenkins.war to start the Jenkins server.
6. **Access Web UI**: Open a web browser and go to http://localhost:8080 to access the Jenkins setup wizard.

Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087

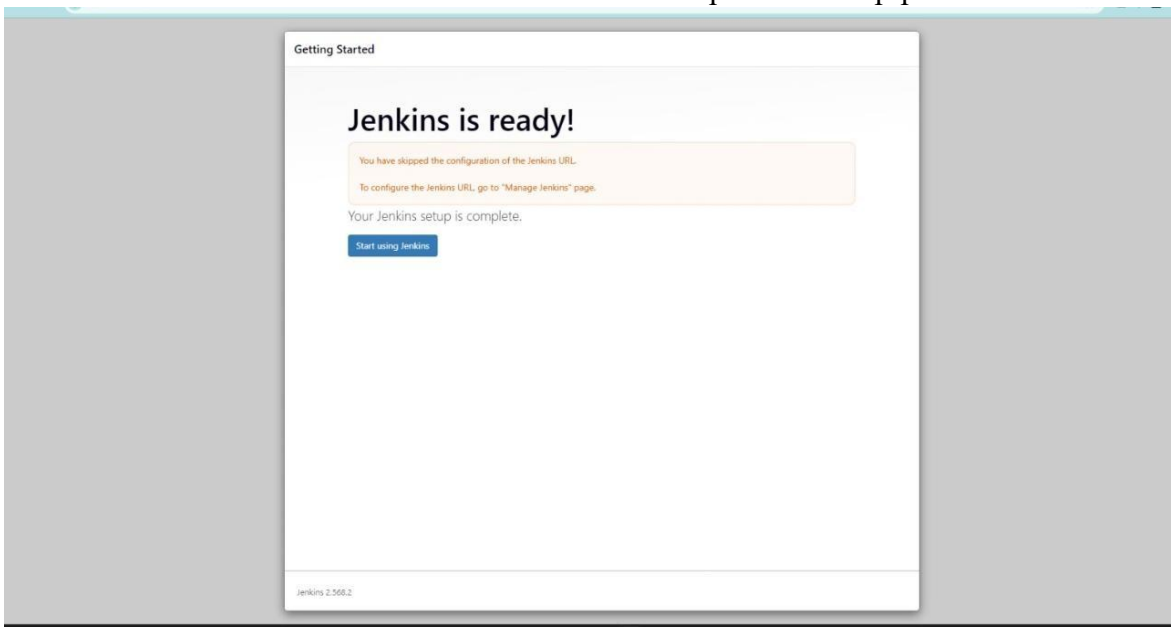




Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087

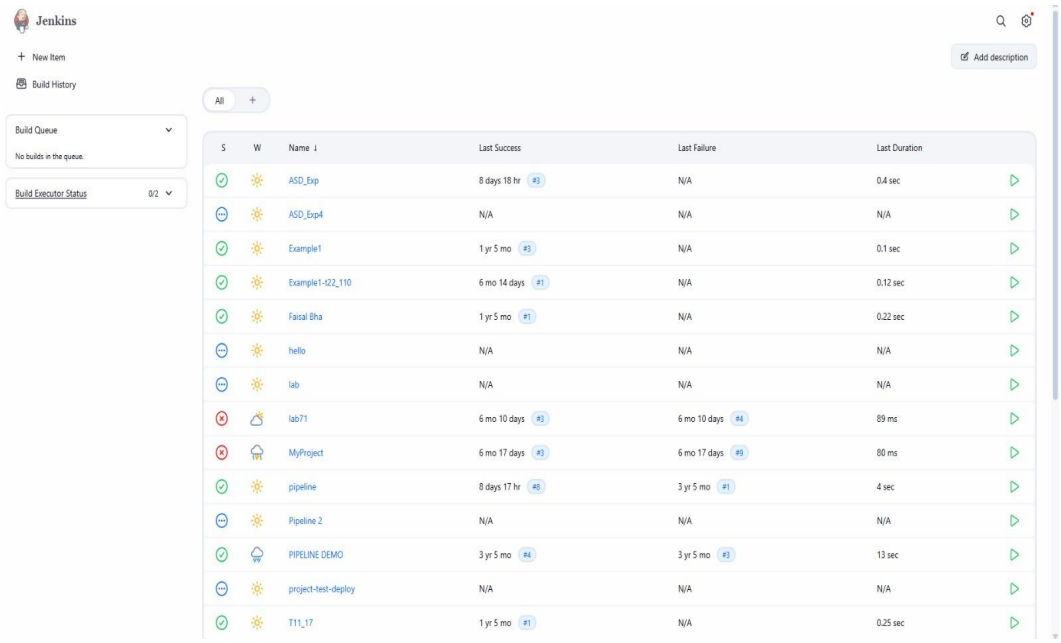
7. **Unlock Jenkins**: Retrieve the initial admin password from the terminal and enter it in the setup wizard.
8. **Customize Plugins**: Choose recommended or custom plugins to install during setup.
9. **Create Admin User**: Set up an admin user account with credentials.

10. **Jenkins URL**: Confirm the Jenkins URL and complete the setup process.



11. **Dashboard**: Log in with the admin credentials to access the Jenkins dashboard.

Name: Nikhil Rajani
Batch: T21
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Conclusion: By this assignment we learned about Continuous integration and way to install Jenkins.

Name: Nikhil Rajani
Batch: T21
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AIM: To build pipeline using Jenkins.
THEORY:

What is a Pipeline in Jenkins?

A Jenkins Pipeline is a set of automated processes that help developers andDevOps teams build, test, and deploy code in a continuous integration/continuousdelivery (CI/CD) environment. In simple terms, it's a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline:

• Automates CI/CD processesProvides better visibility into the build process

• Offers version control of build logic (Pipeline as Code)

• Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines:

There are two main types of syntax used to define Jenkins pipelines:

1. Declarative Pipeline

Simpler and more structured syntaxDesigned for users who want an easy-to-read format Introduced later to make pipelines more readable and maintainable

Key Features:

- Uses pipeline {} block
- Organized into stages and steps
- Easier error handling
- Better validation in Jenkins UI

2. Scripted Pipeline

- More flexible and powerful
- Written in Groovy-based syntax
- Gives full control over the build process
- Suitable for advanced users

Key Features:

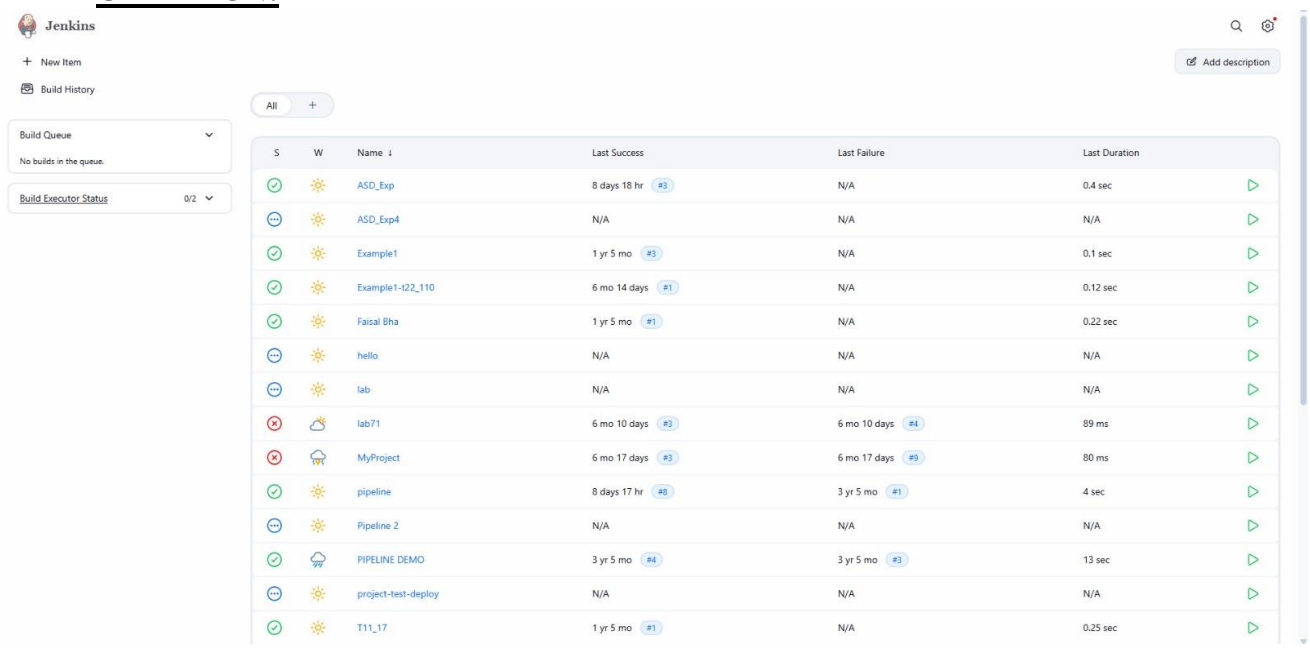
- Uses node {} block
- Everything is written as code

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Batch: T21
Roll No.: 2401087

- Allows complex scripting and logic
- Less user-friendly for beginners

1.Declarative Pipeline

CREATION:

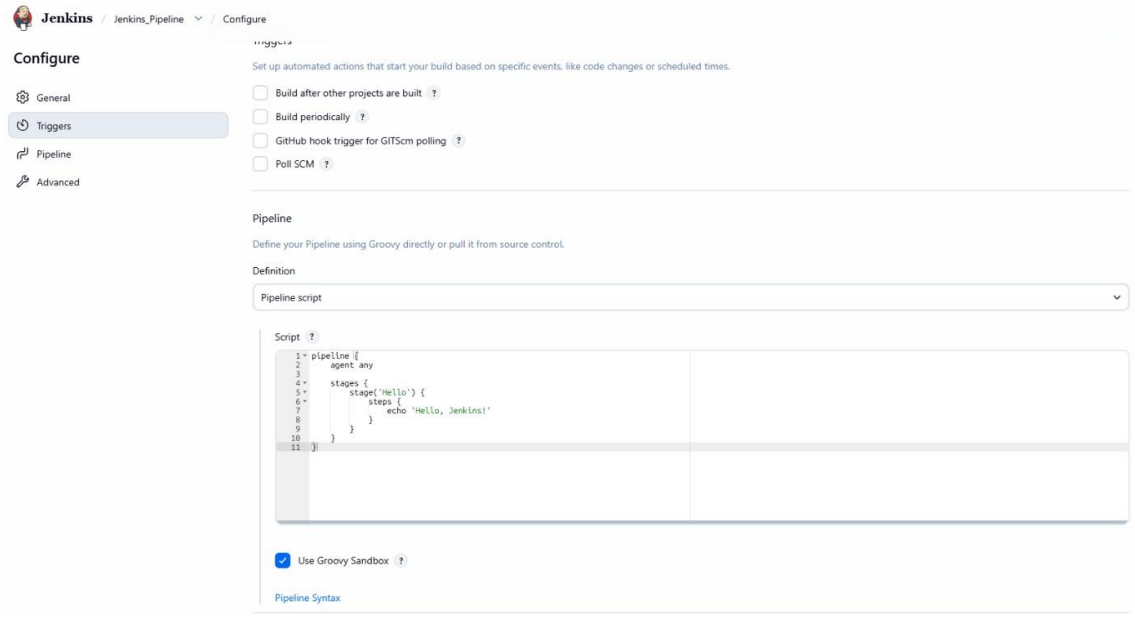


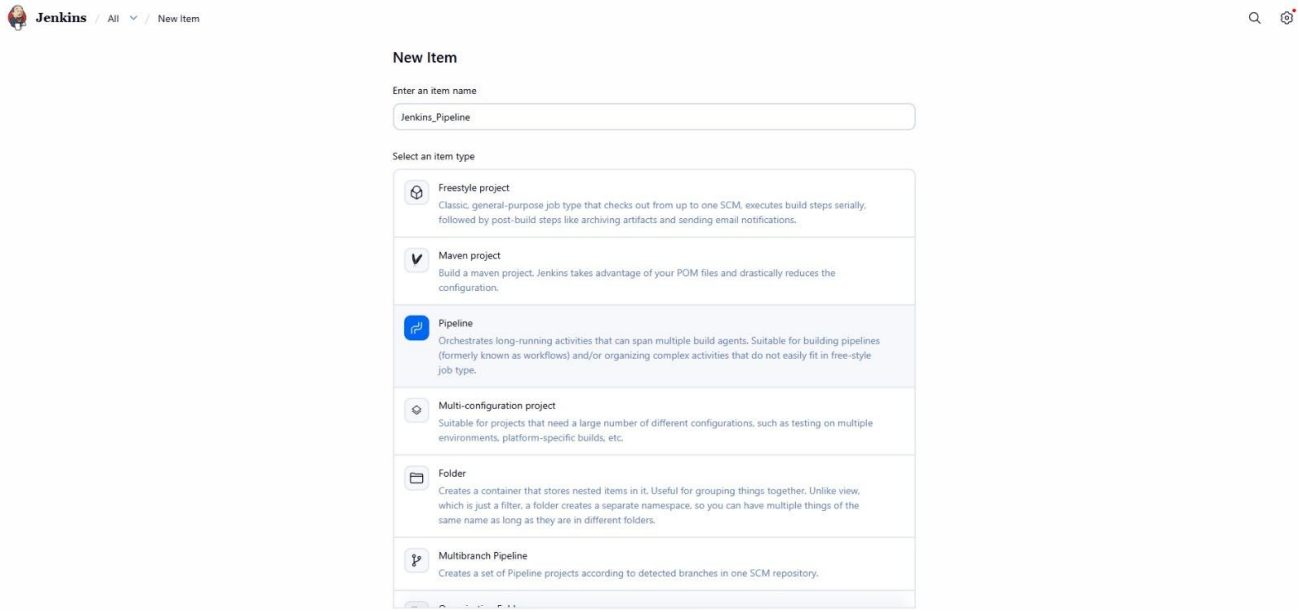
CODE:

```
pipeline {
agent any

    stages { stage('Hello')
{ steps { echo
'Hello, Jenkins!'
            }
        }
    }
}
```

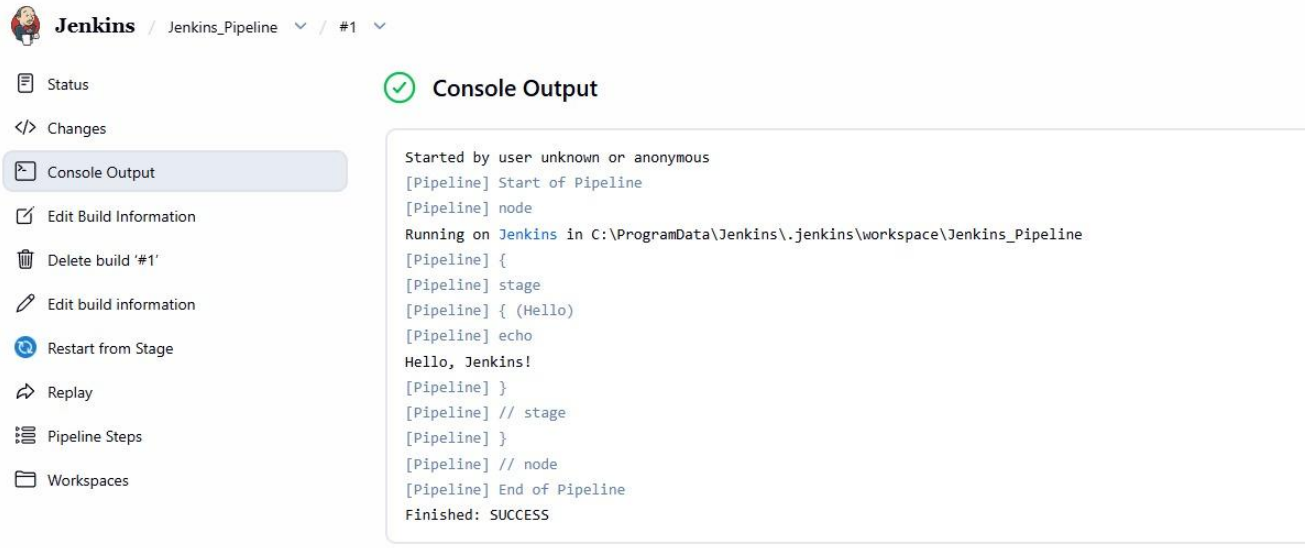
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Batch: T21
Roll No.: 2401087



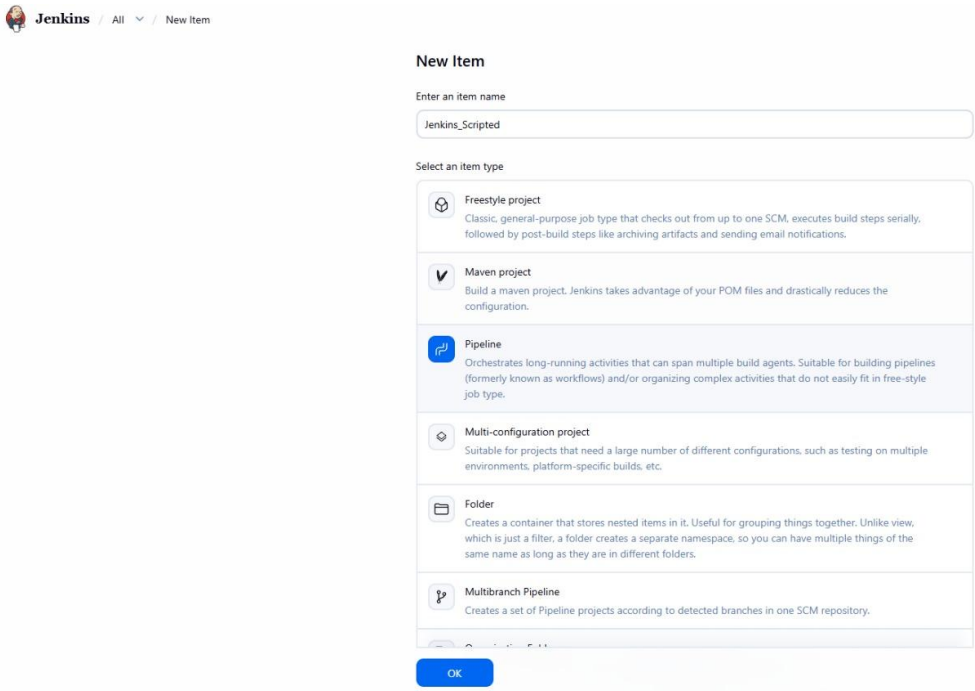


Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087

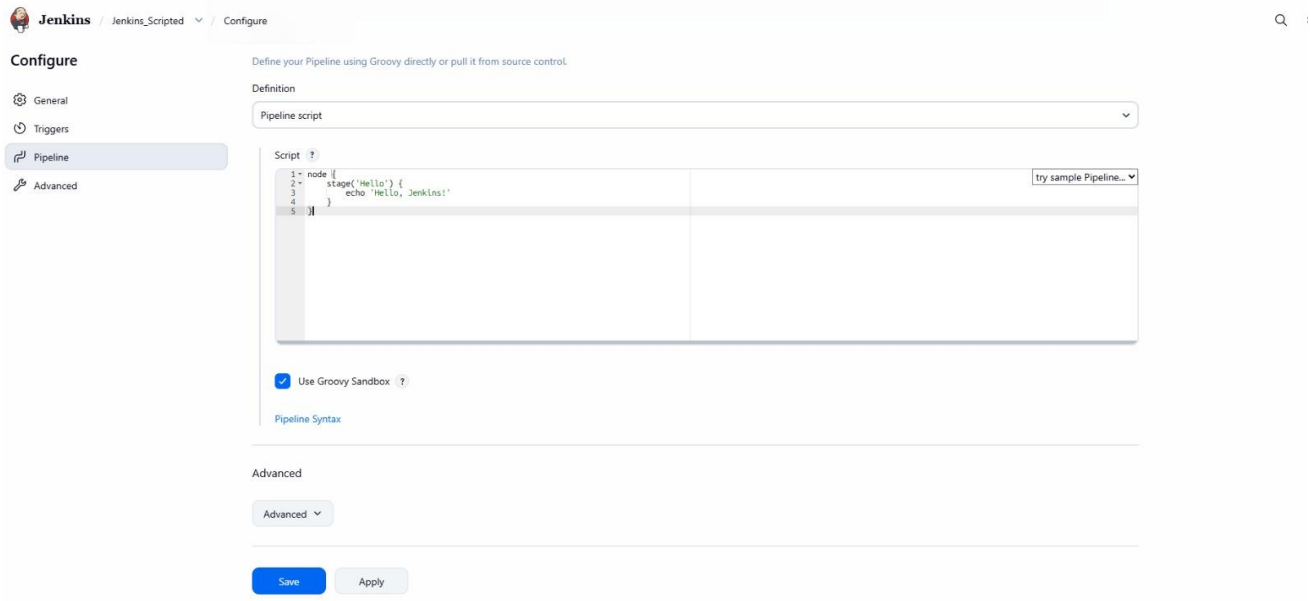
OUTPUT:



SCRIPTED CREATION :



Name: Nikhil Rajani
Batch: T21
Roll No.: 2401087





CONCLUSION: LO1, LO3 in this assignment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows